Categories, Proofs, and Processes: Chapter 3 Limits

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1 Feb 2024

3 Limits and Colimits

3.1 Equalisers

3.1.1 Motivation

In **Set**, we can define equality between functions f, g when $\forall x \in X, f(x) = g(x)$, but we might be interested in situations where where $f(x) = g(x)$ for $x \in Y \subsetneq X$.

For example, $X = \{0,1\}^*$, but only a subset G could encode graphs using this alphabet.

3.1.2 Definition

Definition 3.1 (Equaliser, Def. 38, pg. 25)**.** Consider a pair of parallel arrows $A \underset{g}{\overset{f}{\rightrightarrows}} B$. An ***equaliser*** of (f, g) is an arrow $e : E \to A$ such that

$$f \circ e = g \circ e$$

and for any $h : D \to A$ such that $f \circ h = g \circ h$, there is a unique $\hat{h} : D \to E$ so that

$$h = e \circ \hat{h}$$

Diagrammatically,

$$\begin{tikzcd} E \arrow[r, "e"] & A \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & B \\ D \arrow[u, dashed, "\hat{h}"] \arrow[ur, "h"'] & & \end{tikzcd}$$

Note 3.2. A diagram commutes if any two paths with a common source and target, with at least one of the paths at least length > 1, are equal.

Example 3.3. In **Set**, the equaliser of f, g is given by the inclusion

$$\{x \in A | f(x) = g(x)\} \hookrightarrow A$$

For $f, g : \mathbb{R} \to \mathbb{R}$ where

$$\forall x, f(x) = ax^2 + bx + c, g(x) = 0$$

the equaliser is the roots of $f(x)$ if they exist. ▲

3.1.3 Results

Proposition 3.4. *Any equaliser in any category is monic.*

Proof.

$$\begin{tikzcd}
E \arrow[r, "e"] & A \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & B \\
D \arrow[u, shift left, "x"] \arrow[u, shift right, "y"'] \arrow[ur, "h"'] & &
\end{tikzcd}$$

Suppose $e \circ x = e \circ y$, then it is equal to h. Then

$$f \circ h = f \circ e \circ x = g \circ e \circ x = g \circ h$$

There is a unique map $h' : D \to E$ such that $e \circ h' = h$. But since

$$e \circ x = h \wedge e \circ y = h \implies x = y = h'$$

by uniqueness, so e is monic. □

Definition 3.5. Coequaliser in $\mathcal{C}$ is an equaliser in $\mathcal{C}^{op}$.

Proposition 3.6. *Any coequaliser in any category is epic.*

Proposition 3.7. *Equalisers and coequalisers are unique up to a unique isomorphism.*

Proof. Suppose

$$\begin{tikzcd}
E_1 \arrow[r, "e_1"] & A \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & B \\
E_2 \arrow[ur, "e_2"'] & &
\end{tikzcd}$$

Thus there are unique maps

$$u_1 : E_2 \to E_1, \quad e_2 = e_1 \circ u_1$$
$$u_2 : E_1 \to E_2, \quad e_1 = e_2 \circ u_2$$

Then identities exist

$$1_{E_1} : E_1 \to E_1 := u_1 \circ u_2$$
$$1_{E_2} : E_2 \to E_2 := u_2 \circ u_1$$

□

3.2 Pullbacks

3.2.1 Definitions

Definition 3.8. Given a pair of morphisms

$$A \xrightarrow{f} C \xleftarrow{g} B$$

(called a cospan), we define an (f, g)-cone to be

$$A \xleftarrow{p} D \xrightarrow{q} B$$

such that the following diagram commutes.

$$\begin{tikzcd} D \arrow[r, "q"] \arrow[d, "p"'] & B \arrow[d, "g"] \\ A \arrow[r, "f"'] & C \end{tikzcd}$$

Definition 3.9 (Pullback, Def. 35, pg. 23)**.** The ***pullback*** of f along g is an (f, g)-cone such that for any other (f, g)-cone $A \xleftarrow{p} D \xrightarrow{q} B$, there is a unique $h : D' \to D$ such that $p' = p \circ h, q' = q \circ h$. Diagrammatically,

$$\begin{tikzcd} D' \arrow[dr, dashed, "h"] \arrow[drr, bend left, "q'"] \arrow[ddr, bend right, "p'"'] & & \\ & D \arrow[r, "q"] \arrow[d, "p"'] & B \arrow[d, "g"] \\ & A \arrow[r, "f"'] & C \end{tikzcd}$$

Note 3.10. Pullback often denoted as

$$A \times_C B$$

Example 3.11. In **Set**, pullback is subset of cartesian product. Take a function $f : A \to B$ and subset $V \subseteq B$. Consider

$$f^{-1}(V) = \{a \in A | f(a) \in V\} \subseteq A$$

Consider

$$\begin{tikzcd} f^{-1}(V) \arrow[r, "\bar{f}"] \arrow[d, hook] & V \arrow[d, hook] \\ A \arrow[r, "f"'] & B \end{tikzcd}$$

So $f^{-1}(V)$ is a pullback, unique up to a unique isomorphism. ▲

Proposition 3.12. *Pullbacks are unique up to a unique isomorphism.*

3.2.2 Results

Proposition 3.13. *If a category $\mathcal{C}$ has binary products and equalisers, then it has pullbacks.*

Proof. Given the cospan

$$A \xleftarrow{p} C \xrightarrow{q} B$$

we want to construct its pullback using products and equalisers. Consider the product

$$A \xleftarrow{\pi_1} A \times B \xrightarrow{\pi_2} B$$

Form an equaliser $e : E \to A \times B$. The following illustrative diagram does not commute

$$\begin{tikzcd}
E \arrow[dr, "e"] \arrow[drr, bend left, "\pi_2 \circ e"] \arrow[ddr, bend right, "\pi_1 \circ e"'] & & \\
& A \times B \arrow[r, "\pi_2"] \arrow[d, "\pi_1"'] & B \arrow[d, "g"] \\
& A \arrow[r, "f"'] & C
\end{tikzcd}$$

We want to show $A \xleftarrow{\pi_1 \circ e} E \xrightarrow{\pi_2 \circ e} B$ is a pullback, since e might not be unique. Assume the following diagrams commute

$$\begin{tikzcd}
D \arrow[r, "h"] \arrow[d, "k"'] & B \arrow[d, "g"] \\
A \arrow[r, "f"'] & C
\end{tikzcd}
\qquad\qquad
\begin{tikzcd}
D \arrow[rr, "h"] \arrow[dr, dashed, "\langle k,h \rangle"] \arrow[dd, "k"'] & & B \\
& A \times B \arrow[ur, "\pi_2"'] \arrow[dl, "\pi_1"] & \\
A & &
\end{tikzcd}$$

Then

$$f \circ \pi_1 \circ \langle k, h \rangle = g \circ \pi_2 \langle k, h \rangle$$

Combine this with the original equaliser and we have

$$\begin{tikzcd}
E \arrow[r] & A \times B \arrow[r, shift left, "f \circ \pi_1"] \arrow[r, shift right, "g \circ \pi_2"'] & C \\
D \arrow[u, dashed, "u"] \arrow[ur, "\langle k,h \rangle"'] & &
\end{tikzcd}$$

Unique map u, thus $A \xleftarrow{\pi_1 \circ e} E \xrightarrow{\pi_2 \circ e} B$ is a pullback. □

3.3 Summary

Constructs follow similar pattern

1. Determine shape of a diagram
2. Highlight an object and arrows in that diagram
3. Stipulate unique map to (from) that object from (to) all other objects with the same ”composition” properties

Note 3.14. We are exhibiting a universal (mapping) property where the unique map is a ”unique morphism”.

3.4 Limits

Instead of taking a diagram and highlighting an object, we could have started with a smaller diagram, introduced a new object, and describe its relationship to that diagram.

Example 3.15. Binary products coming from the starting category $\mathcal{J}$ which is the discrete category of 2 objects.

$$\mathcal{J} \qquad \bullet \quad \bullet$$

$$\begin{tikzcd} & C \arrow[dl, "\pi_0"'] \arrow[dr, "\pi_1"] & \\ D_0 & & D_1 \end{tikzcd}$$

▲

Example 3.16. Pullbacks coming from the starting category $\mathcal{J}$ where

$$\mathcal{J} \qquad 0 \xrightarrow{\alpha} 1 \xleftarrow{\beta} 2$$

$$\begin{tikzcd} C \arrow[r] \arrow[d] & D_2 \arrow[d, "D_\beta"] \\ D_0 \arrow[r, "D_\alpha"'] & D_1 \end{tikzcd}$$

▲

Example 3.17. Unary products come from the single object category $\mathbb{1}$. ▲

Example 3.18. Terminal objects come from the empty category. ▲

Definition 3.19 (Diagram, Def. 72, pg. 49)**.** Let $\mathcal{C}$ be a category and $\mathcal{J}$ be an "index" category. A diagram of shape/type $\mathcal{J}$ is a functor

$$D : \mathcal{J} \to \mathcal{C}$$

So we can think of objects as indices $i, j, ...$ and $D_i := D(i)$, evaluated on object i.

Definition 3.20 (Cone, pg. 50)**.** A ***cone*** to a diagram D consists of an object C in $\mathcal{C}$ and a family of arrows in $\mathcal{C}$

$$c_j : C \to D_j$$

, one for each object $j \in \mathrm{Ob}(\mathcal{J})$ such that for each arrow $\alpha : i \to j$ in $\mathcal{J}$, the following diagram commutes

$$\begin{tikzcd} C \arrow[r, "c_j"] \arrow[d, "c_i"'] & D_j \\ D_i \arrow[ur, "D_\alpha"'] & \end{tikzcd}$$

A morphism of cones $\theta : (C, c_j) \to (C', c'_j)$ is an arrow θ in $\mathcal{C}$ making each of the following triangles commute

$$\begin{tikzcd} C \arrow[r, "\theta"] \arrow[dr, "c_j"'] & C' \arrow[d, "c'_j"] \\ & D_j \end{tikzcd}$$

Note 3.21. See the limit as a "gatekeeper" to the rest of the cones.

$$\begin{tikzcd}
 & W \arrow[d] \arrow[ddl, bend right] \arrow[ddr, bend left] & \\
 & \lim_I J \arrow[dl] \arrow[dr] & \\
J\ i \arrow[rr, "J\ f"'] & & J\ j
\end{tikzcd}$$

Fact 3.22. Functors preserve all commutative diagrams and all cones.

Example 3.23. For $\mathcal{J} := A \overset{f}{\underset{g}{\rightrightarrows}} B$, we have a cone where the following commute.

$$\begin{tikzcd}
C \arrow[r, "c_B"] \arrow[d, "c_A"'] & D_B \\
D_A \arrow[ur, "D_f"'] &
\end{tikzcd}
\qquad
\begin{tikzcd}
C \arrow[r, "c_B"] \arrow[d, "c_A"'] & D_B \\
D_A \arrow[ur, "D_g"'] &
\end{tikzcd}$$

thus

$$c_B = D_f \circ c_A = D_g \circ c_A$$

Then if we have a morphism of cones so a similar diagram to above commutes, we also have

$$c'_B = D_f \circ c'_A = D_g \circ c'_A$$

We have the diagram

$$\begin{tikzcd}
C' \arrow[r, "c_A"] & D_A \arrow[r, shift left, "D_f"] \arrow[r, shift right, "D_g"'] & D_B \\
C \arrow[u, "u"] \arrow[ur, "h"'] & &
\end{tikzcd}$$

We still don't have an equaliser yet since u is not necessarily unique. ▲

Definition 3.24 (Limit, pg. 50)**.** A ***limit*** for a diagram $D : \mathcal{J} \to \mathcal{C}$ is a terminal object in **Cone**(D), which is the category of cones in D. A finite limit is a limit where the category $\mathcal{J}$ in the diagram D is a finite category.

Definition 3.25. By dualising, a colimit is an initial object in the category of cocones.

Note 3.26. For particular categories, limits and colimits may not exist.

Definition 3.27. If $\mathcal{C}$ has limits for all $\mathcal{J}$ being finite then it is finitely complete.

Definition 3.28. If $\mathcal{C}$ has limits for all small $\mathcal{J}$ it is complete.

Proposition 3.29. *$\mathcal{C}$ is finitely complete iff it has finite products and equalisers.*

3.5 Preservation

Let $F : \mathcal{C} \to \mathcal{E}$ be a functor. We can compose diagrams with functors. i.e. if $D : \mathcal{J} \to \mathcal{C}$, then $F \circ D : \mathcal{J} \to \mathcal{E}$ is a diagram of shape $\mathcal{J}$ in $\mathcal{E}$.

Cones are also subject to functors i.e. if $(C, (c_1, ...))$ is a cone to D in $\mathcal{C}$, then $(FC, (Fc_1, ...))$ is a cone to FD in $\mathcal{E}$.

Definition 3.30. A functor $F : \mathcal{C} \to \mathcal{E}$ preserves limits of shape $\mathcal{J}$ whenever the cone $(C, (c_1, ...))$ to D is a limit, then the cone $(FC, (Fc_1, ...))$ is a limit to FD.

Fact 3.31. Covariant Hom-functors $\mathcal{C}(A, -)$ preserve limits.

Contravariant Hom-functors $\mathcal{C}(-, B)$ preserve colimits.

3.6 Exponentials

3.6.1 Motivating Example

Consider a Boolean function $f : \{0,1\}^n \to \{0,1\}$. Describe it as a tuple of length 2^n, where elements are indexed by the input $x \in \{0,1\}^n$.

Since there are 2^n possible inputs, and 2 possible assignments to each input, there are 2^{2^n} possible functions.

More generally for two finite sets A, B, the set of functions $f : A \to B$ has cardinality $|B|^{|A|}$.

Let Boolean functions $f(x, \bar{x})$ where $x \in \{0,1\}, \bar{x} \in \{0,1\}^{n-1}$.

If we fix a particular value of x, we have two possible functions

$$f_0 : \{0,1\}^{n-1} \to \{0,1\} \quad \text{if } x = 0$$

$$f_1 : \{0,1\}^{n-1} \to \{0,1\} \quad \text{if } x = 1$$

f_0, f_1 belong to the set of functions $\{0,1\}^{n-1} \to \{0,1\}$ of cardinality $2^{2^{n-1}}$.

If we vary x we have a map

$$\tilde{f} : \{0,1\} \to \{0,1\}^{2^{n-1}}$$

From x to the functions $f_x : \{0,1\}^{n-1} \to \{0,1\} := x \mapsto f(x, \bar{x})$, which is uniquely determined by

$$\tilde{f}(x)(\bar{x}) = f_x(\bar{x}) = f(x, \bar{x})$$

3.6.2 Evaluation

In general, for a function $f : A \times B \to C$, we have the map $\tilde{f} : A \to C^B$ such that

$$\tilde{f}(a)(b) = f(a, b)$$

uniquely.

That is, we have an isomorphism of hom-sets:

$$\mathbf{Set}(A \times B, C) \cong \mathbf{Set}(A, C^B)$$

i.e. a bijection of the functions in each set.

We have an evaluation function

$$\texttt{eval} : C^B \times B \to C := (g, b) \mapsto g(b)$$

for $g \in C^B$.

Evaluation has a **universal mapping property** Given any set A and function

$$f : A \times B \to C$$

there is a unique function

$$\tilde{f} : A \to C^B$$

such that

$$\texttt{eval} \circ (\tilde{f} \times 1_B) = f$$

i.e. for sets,

$$\texttt{eval}(\tilde{f}(a), b) = f(a, b)$$

Diagrammatically,

$$\begin{tikzcd} C^B & & C^B \times B \arrow[r, "\texttt{eval}"] & C \\ A \arrow[u, "\tilde{f}"] & & A \times B \arrow[u, "\tilde{f} \times 1_B"] \arrow[ur, "f"'] & \end{tikzcd}$$

3.6.3 Definition

Definition 3.32 (Exponential, Def. 74, pg. 51)**.** Let $\mathcal{C}$ have binary products. An ***exponential*** of objects B, C of $\mathcal{C}$ consists of an object

$$C^B$$

and arrow

$$\epsilon : C^B \times B \to C$$

s.t. for any object A and arrow

$$f : A \times B \to C$$

there is a unique arrow

$$\tilde{f} : A \to C^B$$

s.t.

$$\epsilon \circ (\tilde{f} \times 1_B) = f$$

Claim 3.33.

$$\mathcal{C}(A \times B, C) \cong \mathcal{C}(A, C^B)$$

Definition 3.34 (CCC, Def. 78, pg. 52)**.** A category with a terminal object, products, and exponentials is called a **Cartesian Closed Category** (CCC).

Fact 3.35. Exponentials are right adjoints to products.